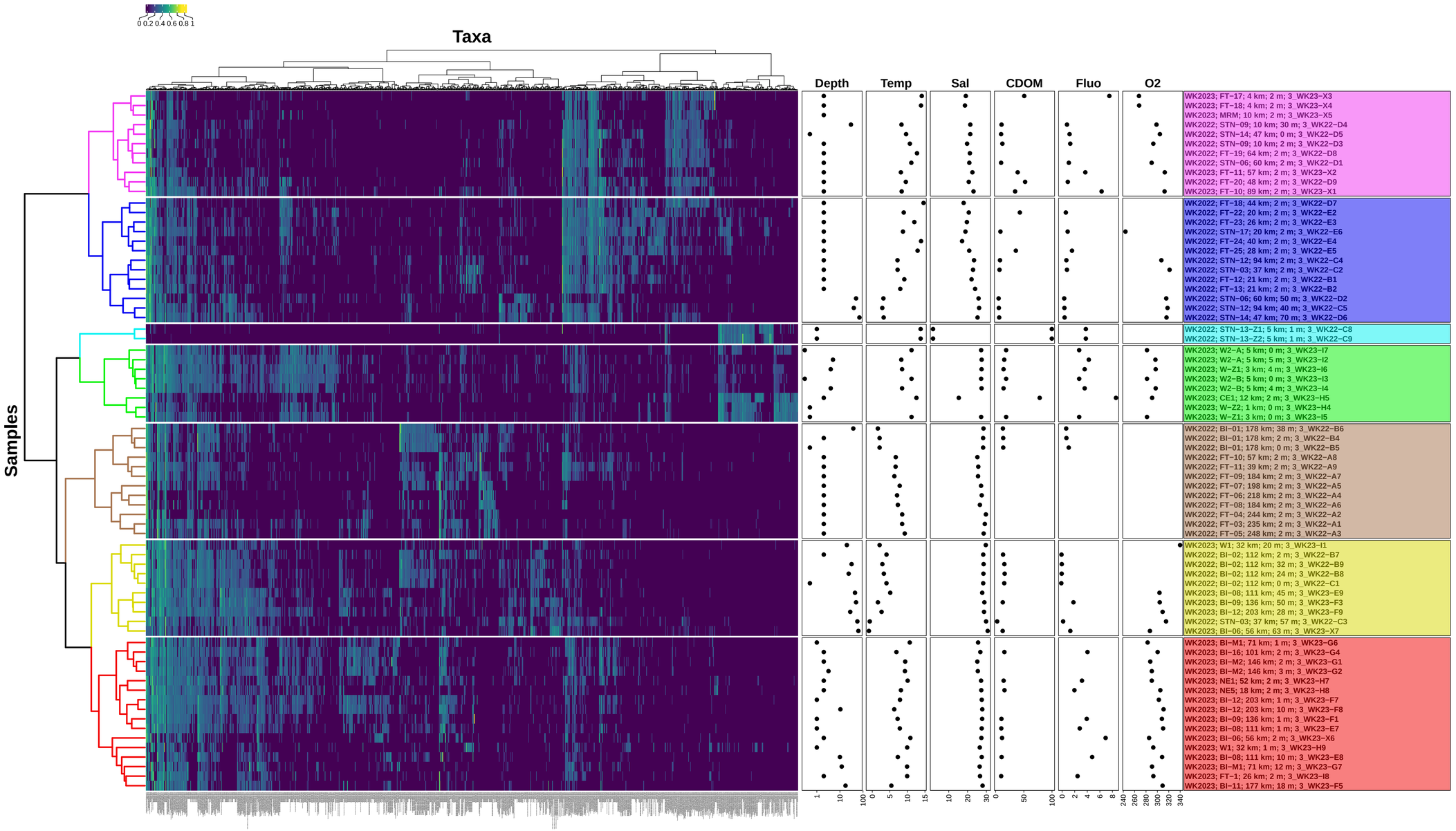
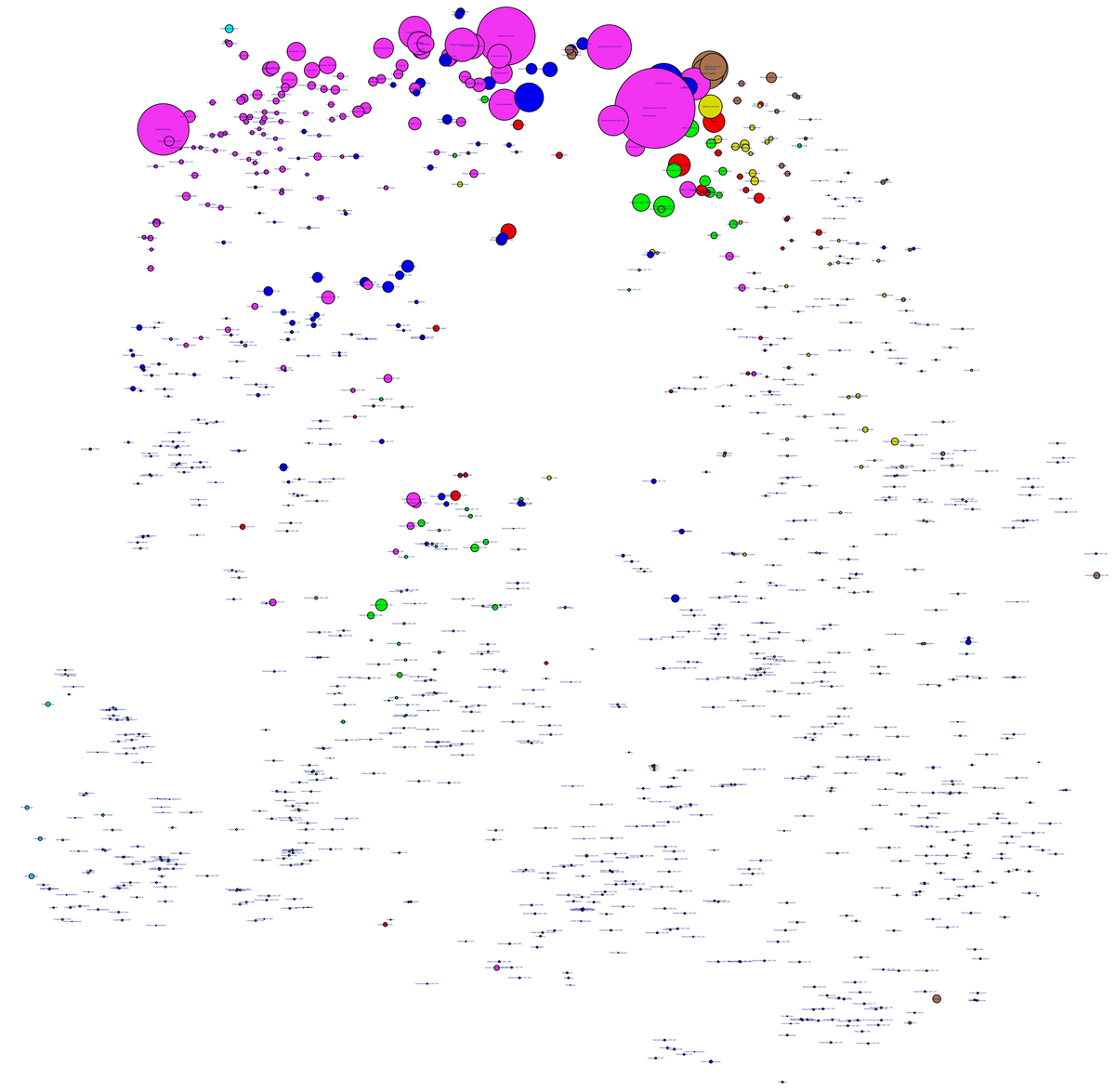
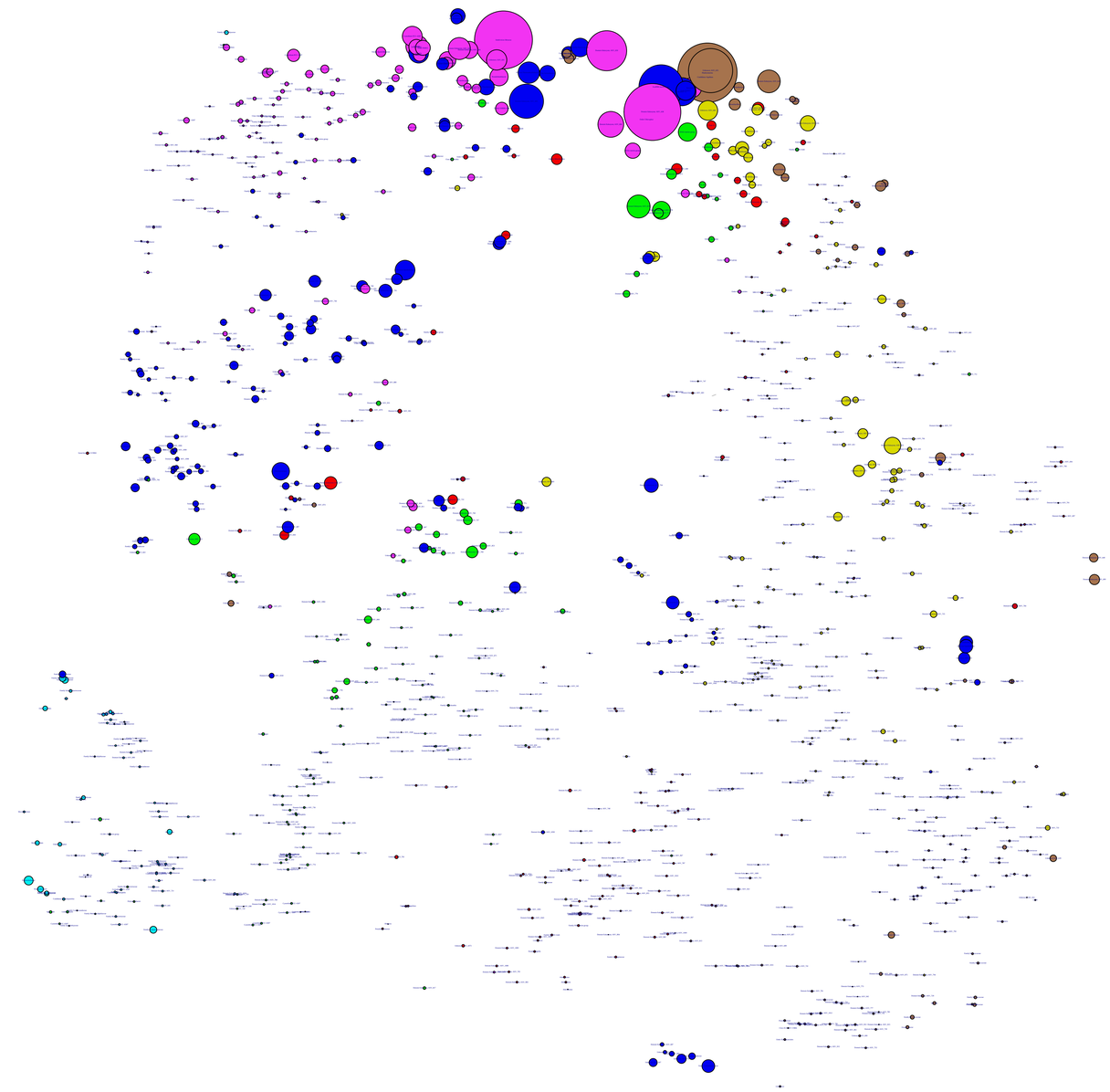
Fifth−root transformed relative abundances of microbial taxa during the 2022 and 2023 James Bay expeditions



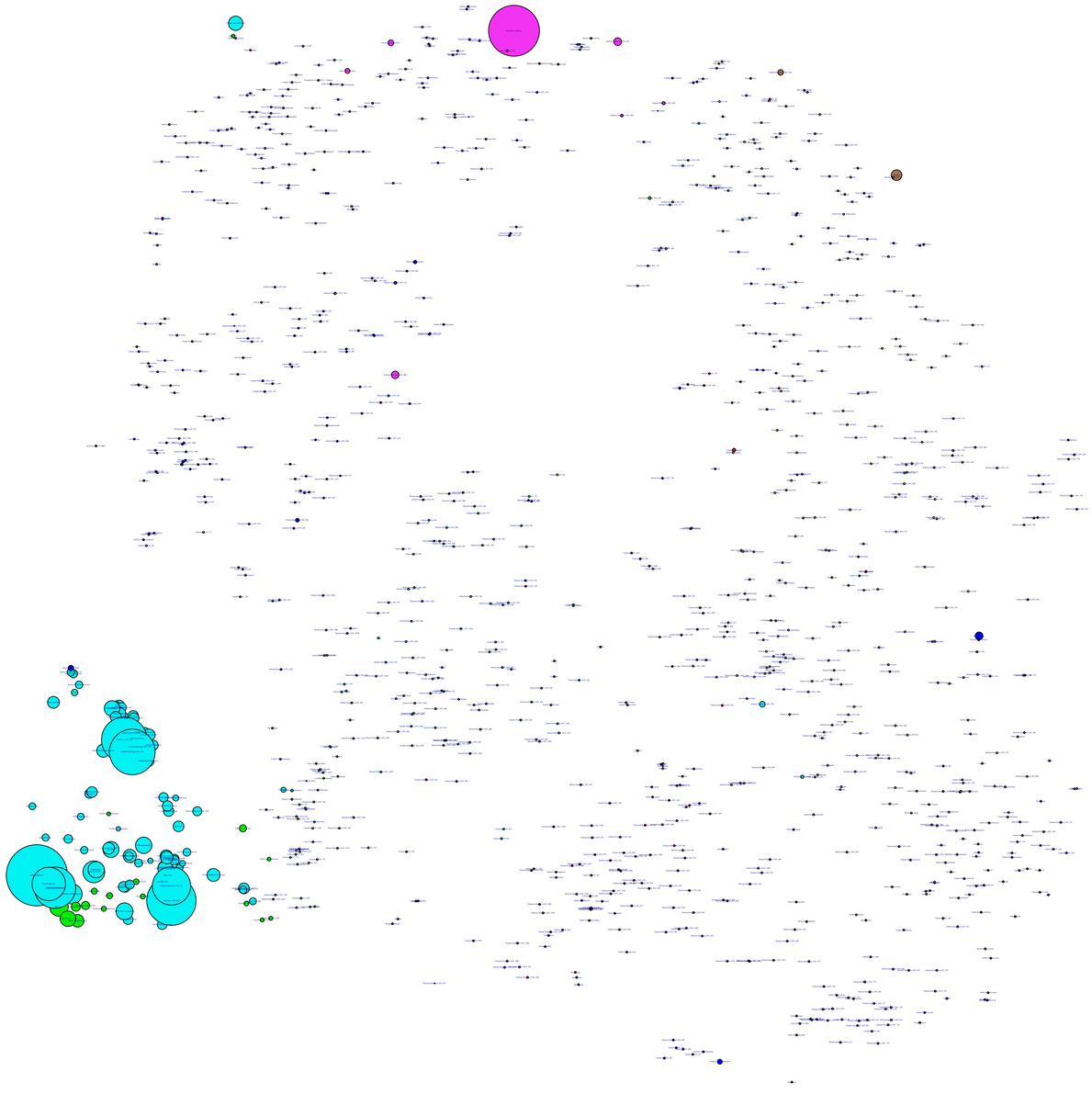
Square−root transformed relative abundances of microbial taxa in Cluster 1 (ASVs are colored by Cluster they are most associated with)



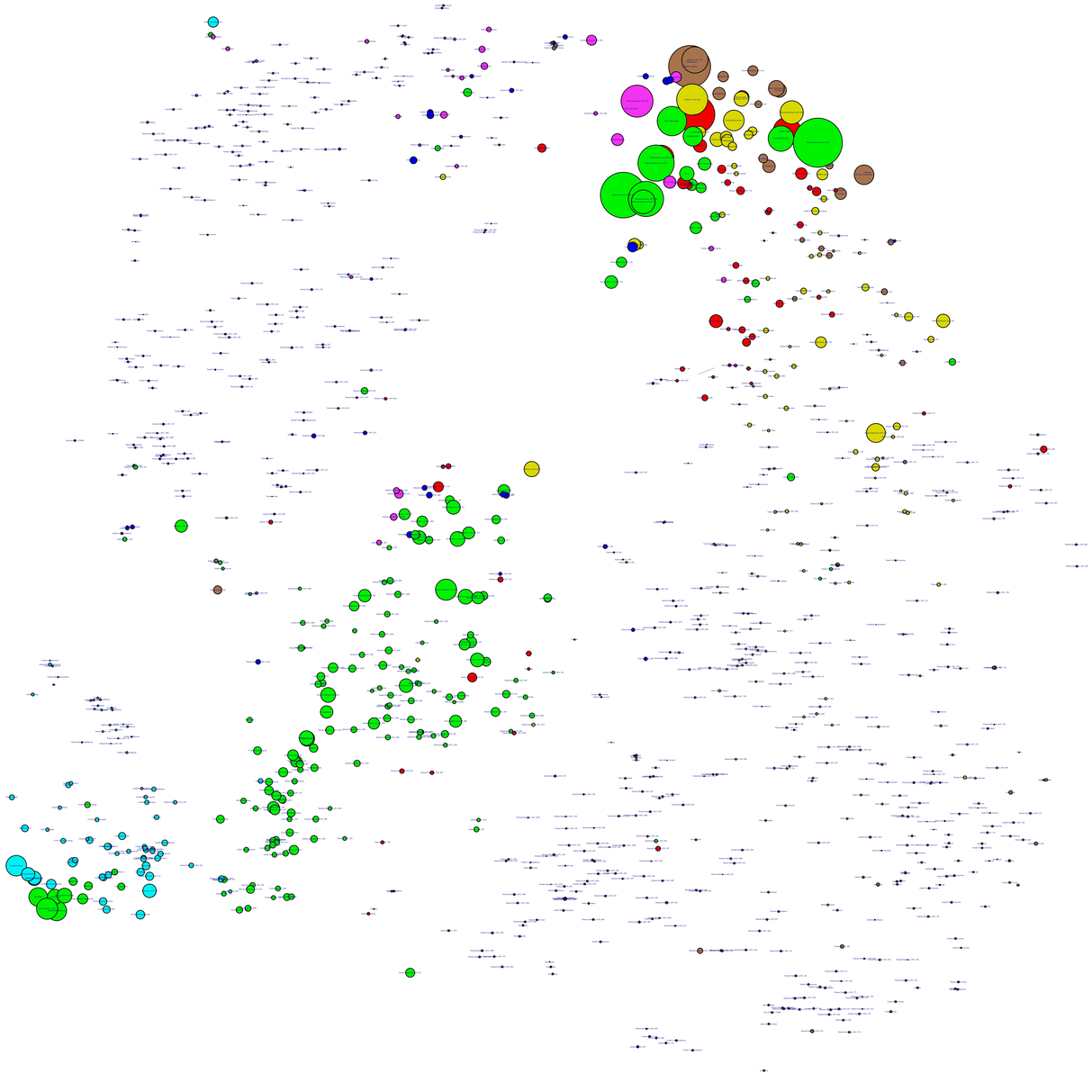
Square−root transformed relative abundances of microbial taxa in Cluster 2 (ASVs are colored by Cluster they are most associated with)



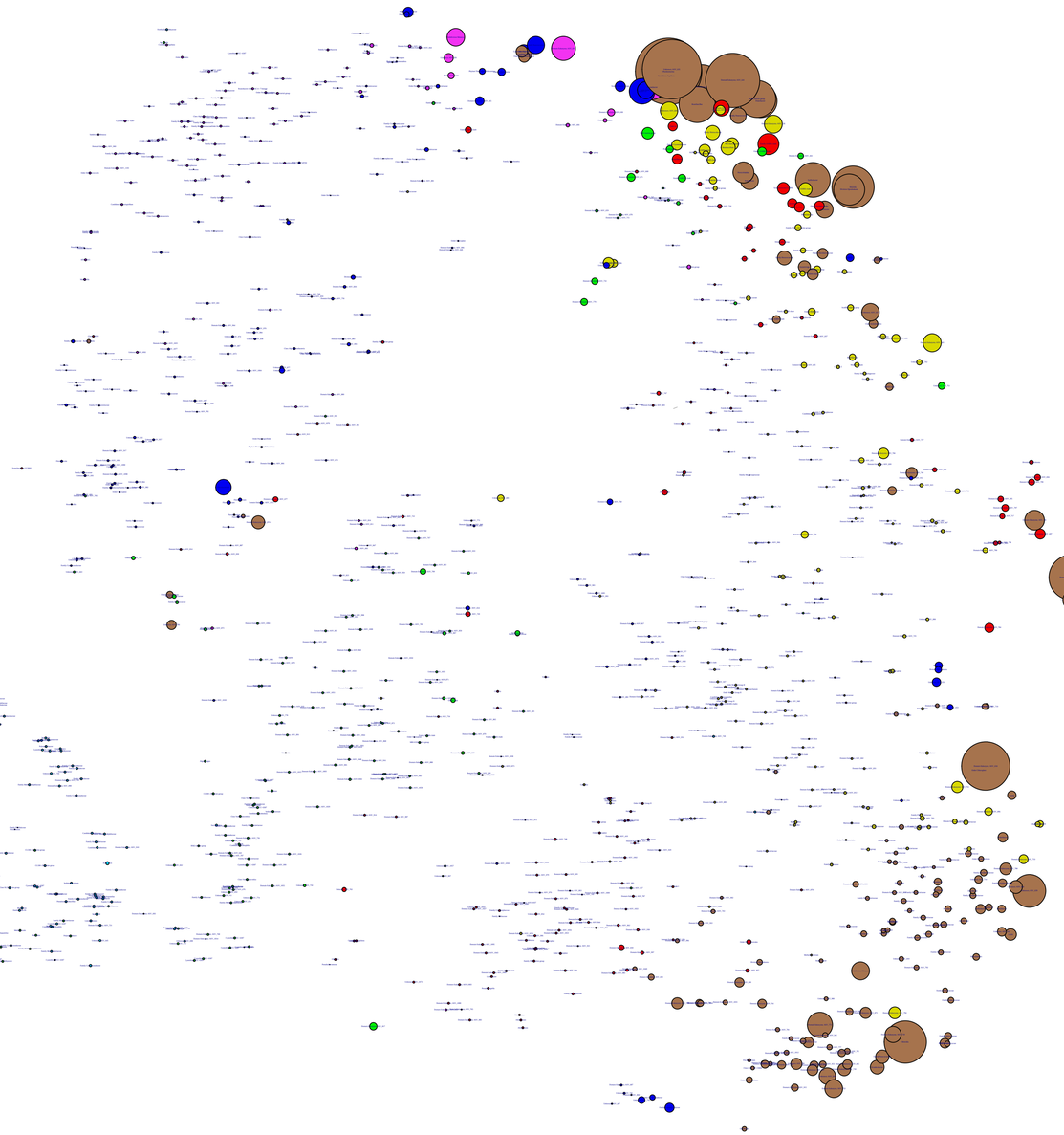
Square−root transformed relative abundances of microbial taxa in Cluster 3 (ASVs are colored by Cluster they are most associated with)



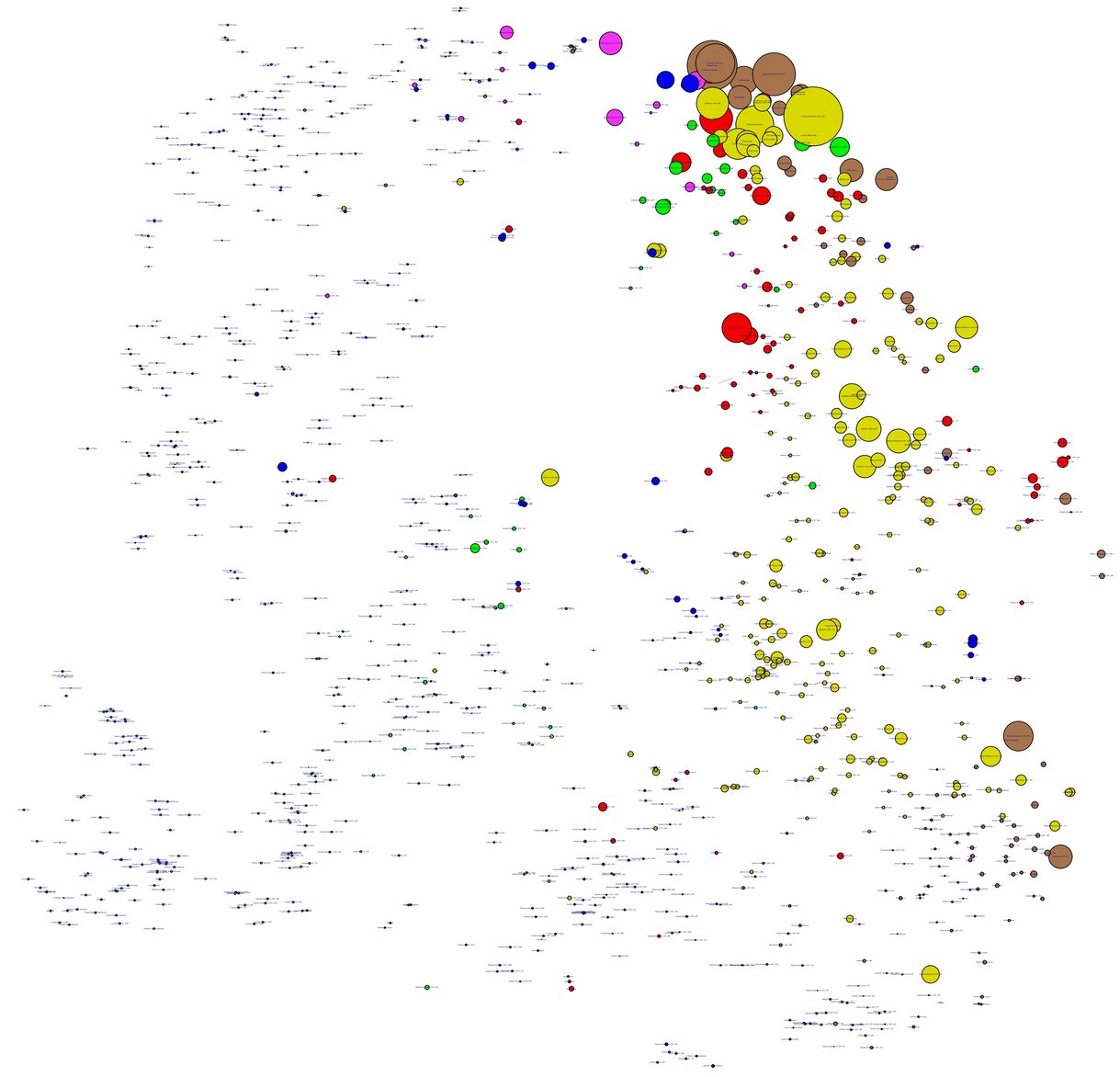
Square−root transformed relative abundances of microbial taxa in Cluster 4 (ASVs are colored by Cluster they are most associated with)

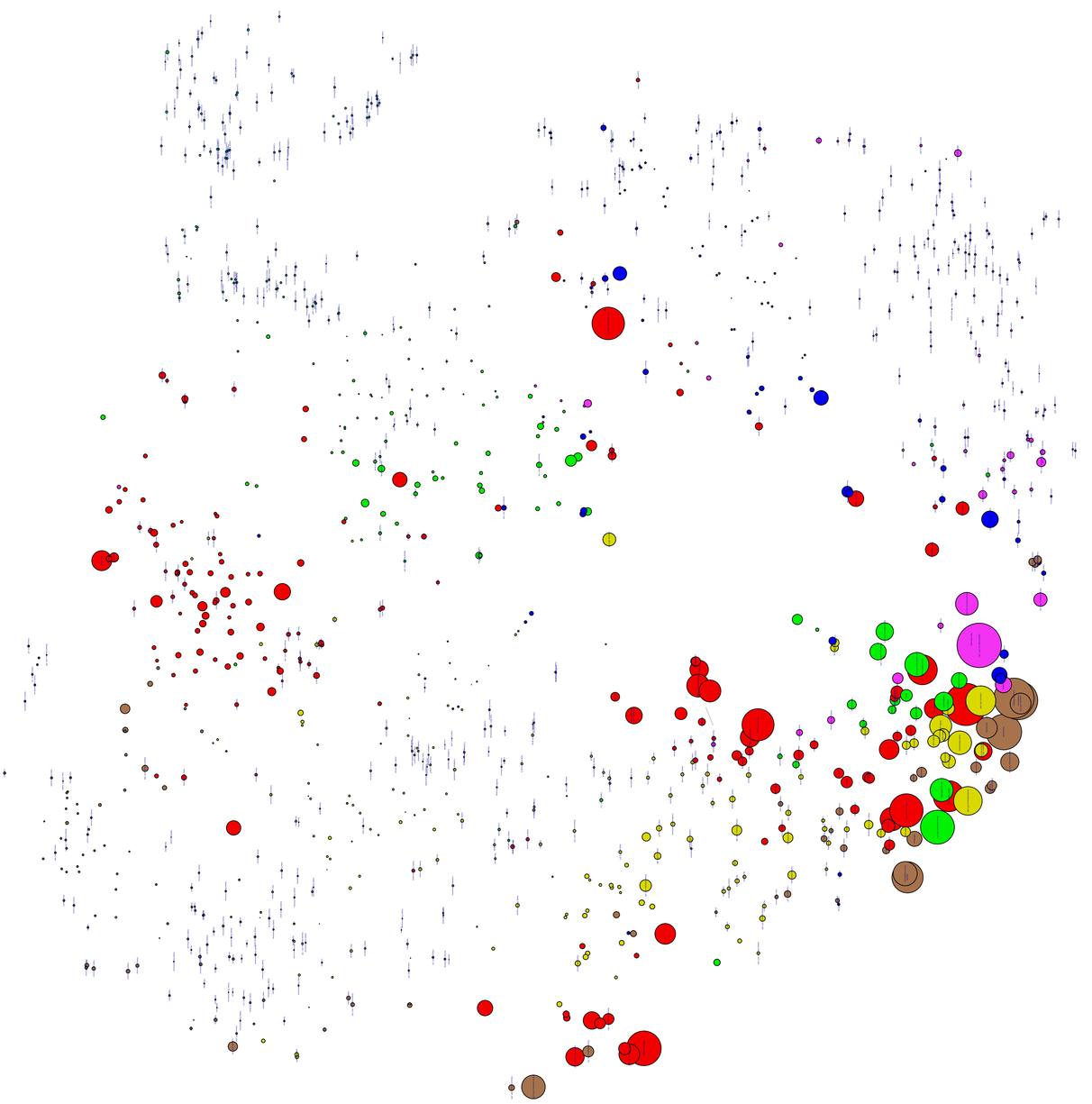


Square−root transformed relative abundances of microbial taxa in Cluster 5 (ASVs are colored by Cluster they are most associated with)



Square−root transformed relative abundances of microbial taxa in Cluster 6 (ASVs are colored by Cluster they are most associated with)





Most abundant identified Cluster−specific ASVs
(>2% mean relative abundance)

Cluster	ASV	Abundance
1	Opisthokonta; Subdivision Metazoa; ASV_647	0.130
1	Opisthokonta; Subdivision Metazoa; ASV_673	0.100
1	Cyanobacteriota; Chloroplast; Order Chloroplast; ASV_122	0.058
1	Cyanobacteriota; Synechococcales; Cyanobium PCC−6307; ASV_220	0.041
1	Actinomycetota; Microtrichales; Ilumatobacter; ASV_56	0.040
1	Pseudomonadota; Pseudomonadales; Family SAR86 clade; ASV_187	0.038
1	Actinomycetota; PeM15; Order PeM15; ASV_302	0.021
2	Actinomycetota; Microtrichales; Sva0996 marine group; ASV_76	0.077
2	Actinomycetota; Micrococcales; ML602J−51; ASV_75	0.030
3	Pseudomonadota; Burkholderiales; Polynucleobacter; ASV_72	0.140
3	Actinomycetota; Frankiales; Family Sporichthyaceae; ASV_123	0.066
3	Pseudomonadota; Burkholderiales; Polynucleobacter; ASV_129	0.043
3	Actinomycetota; Frankiales; Family Sporichthyaceae; ASV_59	0.024
4	Pseudomonadota; Burkholderiales; RS62 marine group; ASV_137	0.034
4	Pseudomonadota; Pseudomonadales; Family SAR86 clade; ASV_18	0.025
5	Pseudomonadota; Rhodobacterales; Planktomarina; ASV_8	0.160
5	Actinomycetota; Micrococcales; Candidatus Aquiluna; ASV_12	0.071
5	Pseudomonadota; Pseudomonadales; Kineobactrum; ASV_3	0.069
5	Opisthokonta; Crustacea; Metridia; ASV_649	0.064
5	Opisthokonta; Crustacea; Metridia; ASV_660	0.064
5	Pseudomonadota; Burkholderiales; RS62 marine group; ASV_83	0.049
5	Verrucomicrobiota; Verrucomicrobiales; Roseibacillus; ASV_13	0.047
5	Pseudomonadota; Rhodobacterales; Sulfitobacter; ASV_11	0.043
5	Bacteroidota; Flavobacteriales; Polaribacter; ASV_4	0.040
5	Opisthokonta; Division Opisthokonta; ASV_672	0.034
6	Pseudomonadota; Pseudomonadales; Family Nitrincolaceae; ASV_2	0.055
6	Pseudomonadota; Pseudomonadales; SUP05 cluster; ASV_9	0.036
6	Thermoproteota; Nitrosopumilales; Candidatus Nitrosopumilus; ASV_14	0.024
6	Pseudomonadota; Pseudomonadales; Pseudohongiella; ASV_43	0.021
7	Pseudomonadota; Pelagibacterales; Clade Ia; ASV_1	0.069
7	Opisthokonta; Division Opisthokonta; ASV_679	0.046
7	Cyanobacteriota; Synechococcales; Synechococcus CC9902; ASV_17	0.043
7	Pseudomonadota; Pseudomonadales; Family SAR86 clade; ASV_6	0.037
7	Bacteroidota; Flavobacteriales; Formosa; ASV_5	0.021

Cluster−specific values of defining environmental parameters

Cluster	Code	Depth (m)	Distance to coast (km)	Temperature (deg C)	Salinity (PSU)	CDOM (mg/m^3)	Fluorescence (mg/m^3)	Shannon diversity
1	FT	4.4 ± 8.5	36.6 ± 29.9	10.7 ± 2.3	20.9 ± 1.4	27.3 ± 18.9	2.8 ± 2.7	3.8 ± 0.5
2	FT	13.8 ± 23.4	42.5 ± 26	8.6 ± 4	21.9 ± 3	14.8 ± 15.3	0.7 ± 0.4	4.3 ± 0.3
3	NISK	1 ± 0	4.9 ± 0.6	13.8 ± 0	1.6 ± 0	99.7 ± 0	3.8 ± 0	3.4 ± 0.2
4	ROS	2.1 ± 2	5.1 ± 3.1	10.2 ± 1.7	25.6 ± 4.5	25.3 ± 23.4	4 ± 2.1	4.7 ± 0.2
5	FT	4.9 ± 10.4	178.4 ± 66.5	6.1 ± 2.7	27.5 ± 1.5	NA	NA	3.8 ± 0.3
6	ROS	32.1 ± 21.6	102.3 ± 50.6	2.4 ± 2	28.9 ± 0.9	12.5 ± 4.6	0.4 ± 0.8	4.4 ± 0.3
7	ROS	4.4 ± 5.1	103.7 ± 60.8	8.6 ± 1.7	26.9 ± 0.9	12.1 ± 2.5	3.8 ± 1.6	4.2 ± 0.6
Kruskal−Wallis p	NA	6.74e−03	3.44e−08	2.76e−06	2.47e−09	2.64e−02	8.35e−05	6.43e−05